

$$C_{\phi} 2^2 \rightarrow -\hat{m}_{H_d} +$$

$$\begin{aligned} & \frac{1}{16 \pi^2} \left(-\frac{1}{2} \sum_p g_1^2 \text{LF}_{1,0} [m_{\tilde{d}}^p] + 3 \overline{y_d^{pr}} y_d^{pr} \text{LF}_{1,0} [m_{\tilde{d}}^r] - \frac{1}{2} \sum_p g_1^2 \text{LF}_{1,0} [m_{\tilde{e}}^p] + \overline{y_e^{pr}} y_e^{pr} \text{LF}_{1,0} [m_{\tilde{e}}^r] + \right. \\ & \left(\overline{y_e^{pr}} y_e^{pr} + \frac{1}{2} \sum_p g_1^2 \right) \text{LF}_{1,0} [m_{\tilde{l}}^p] + \left(3 \overline{y_d^{pr}} y_d^{pr} - \frac{1}{2} \sum_p g_1^2 \right) \text{LF}_{1,0} [m_{\tilde{q}}^p] + \\ & \sum_p g_1^2 \text{LF}_{1,0} [m_{\tilde{u}}^p] - g_1^2 \text{LF}_{1,1,-1} [m_1, \tilde{\mu}] - 3 g_2^2 \text{LF}_{1,1,-1} [m_2, \tilde{\mu}] + \\ & \left. 3 \overline{a_d^{pr}} a_d^{pr} \text{LF}_{1,1,0} [m_{\tilde{d}}^r, m_{\tilde{q}}^p] + \overline{a_e^{pr}} a_e^{pr} \text{LF}_{1,1,0} [m_{\tilde{e}}^r, m_{\tilde{l}}^p] + 3 \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} \text{LF}_{1,1,0} [m_{\tilde{q}}^p, m_{\tilde{u}}^r] \right) \end{aligned}$$